

CSA65 – Generative AI and Large Language Models

Course Objectives

1. To understand the fundamental concepts of Artificial Intelligence, Generative AI, Foundation Models, and Large Language Models.
2. To learn transformer architectures, prompt engineering techniques, and the principles of adapting Large Language Models.
3. To develop Generative AI applications using LLM APIs, Retrieval-Augmented Generation (RAG), and AI agent frameworks.
4. To build intelligent applications integrating text, image, and speech models using modern AI development tools.
5. To understand responsible AI practices, model deployment, security, ethics, and emerging trends in Generative AI.

Course Outcomes

Upon successful completion of the course, students will be able to:

CO1: Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models.

CO2: Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems.

CO3: Develop intelligent applications using Retrieval-Augmented Generation (RAG), vector databases, and introductory AI agent concepts.

CO4: Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.

CO5: Deploy Generative AI applications while evaluating ethical, security, governance, and responsible AI considerations in industrial environments.

Topics

Fundamentals of Generative AI and Large Language Models (9 Hours)

Introduction to Artificial Intelligence – Machine Learning and Deep Learning Overview – Generative AI and Foundation Models – Discriminative and Generative Models – Evolution of Language Models – Transformers and Self-Attention Mechanism – Tokenization and Embeddings – Overview of BERT, GPT, LLaMA, and Gemini Models – Mathematical Foundations: Probability, Neural Networks, Gradient Descent – Applications of Generative AI.

Prompt Engineering and Large Language Models (9 Hours)

Large Language Model Architecture – Pre-training and Fine-tuning Concepts – Transfer Learning – Prompt Engineering Fundamentals – Zero-shot, One-shot, Few-shot and Chain-of-Thought Prompting – Structured Prompting – Function Calling (Introduction) – Parameter-Efficient Fine-Tuning (Overview of LoRA and PEFT) – Using LLM APIs (OpenAI, Gemini and Hugging Face) – Prompt Evaluation and Best Practices.

Retrieval-Augmented Generation and AI Agents (9 Hours)

Embeddings and Semantic Search – Vector Databases (FAISS and ChromaDB) – Retrieval-Augmented Generation (RAG) Architecture – Document Retrieval and Knowledge Augmentation – AI Agents: Concepts and Components – Tool Calling and Agent Workflows (Introduction) – LangChain Basics – Enterprise Applications of RAG and AI Agents – Case Studies.

Multimodal Generative AI and Intelligent Applications (9 Hours)

Text Generation – Text Summarization – Machine Translation – Question Answering Systems – AI Chatbots – Code Generation – Text-to-Image Models – Speech-to-Text – Text-to-Speech – Multimodal AI Concepts – Generative AI Applications in Healthcare, Education, Finance, Software Development, and Engineering.

Deployment, Responsible AI and Future Trends (9 Hours)

Deployment of LLM Applications using FastAPI and Streamlit – Local LLMs using Ollama (Introduction) – Model Optimization Concepts – Responsible AI – AI Ethics – Bias and Fairness – Explainability – AI Security – Prompt Injection and Hallucination (Introduction) – AI Governance – Future Trends in Generative AI.

Text Books

1. **Tanmoy Chakraborty**, *Introduction to Large Language Models: Generative AI for Text*, Wiley India Pvt. Ltd., First Edition, 2024.
2. **Karthikeyan Sabesan, Sivagami sundari, Nilip Dutta**, *Generative AI for Everyone: Deep Learning, NLP and Large Language Models for Creative and Practical Applications*, BPB Publications, First Edition, 2025.

References

1. **Lewis Tunstall, Leandro von Werra, Thomas Wolf**, *Natural Language Processing with Transformers*, O'Reilly Media, 2022.
2. **Denis Rothman**, *Transformers for Natural Language Processing*, Third Edition, Packt Publishing, 2024.
3. **Jay Alammar and Maarten Grootendorst**, *Hands-On Large Language Models*, O'Reilly Media, 2024.
4. **Sebastian Raschka**, *Build a Large Language Model (From Scratch)*, Manning Publications, 2024.
5. **Daniel Jurafsky and James H. Martin**, *Speech and Language Processing*, Third Edition (Online Draft).